

In []: PAWAR SAKSHI MAHENDRA
ROLL NO - 49
PRACTICAL NO - 05

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In [1]: import numpy as np

# define two pairs of vectors
x1 = np.array([1, 1, 1, -1])
y1 = np.array([1, -1])

x2 = np.array([-1, -1, 1, 1])
y2 = np.array([-1, 1])

# compute weight matrix W
W = np.outer(y1, x1) + np.outer(y2, x2)

# define BAM function
def bam(x):
    y = np.dot(W, x)
    y = np.where(y >= 0, 1, -1)
    return y

# test BAM with input
x_test = np.array([1, -1, -1, -1])
y_test = bam(x_test)

# print output
print("Input x: ", x_test)
print("Output y: ", y_test)
```

Input x: [1 -1 -1 -1]
Output y: [1 -1]

In []: